BDA PRACTICAL EXPERIMENTS

Big Data Analytics • Hive & R Programming

A practical-exam-ready guide with aims, commands/programs, expected outputs, execution steps, results, and quick viva revision.

Experiment 1	Hive commands: database and table operations
Experiment 2	R vector arithmetic
Experiment 3	Sorting a vector
Experiment 4	Functions with and without arguments
Experiment 5	Sequence, mean, maximum, and sum

Exam tip: Practise typing each program once without looking at the answer. Then explain the output in your own words.

Experiment 1: Hive Commands

Aim: To explain Hive commands and create a database and table, then describe the table.

Concept: CREATE creates databases/tables; SHOW lists objects; USE selects a database; DESCRIBE shows table columns and data types; DROP removes an object.

Program / Commands

```
-- Start Hive in the lab environment
hive

-- Create and list a database
CREATE DATABASE college;
SHOW DATABASES;

-- Select the database
USE college;

-- Create and list a table
CREATE TABLE student (
    id INT,
    name STRING,
    marks FLOAT
);
SHOW TABLES;

-- Display the table structure
DESCRIBE student;

-- Cleanup (run only after you have finished)
DROP TABLE student;
DROP DATABASE college;
```

Expected Output

```
-- SHOW DATABASES includes:
college

-- SHOW TABLES (after USE college):
student

-- DESCRIBE student:
id       int
name     string
marks    float
```

Procedure

1. Open the Hive terminal or the interface configured by your college.

2. Enter the Hive shell command only if your lab uses that shell; some setups use Beeline.

3. Run the SQL statements in order, one at a time.

4. Check the results of SHOW DATABASES, SHOW TABLES, and DESCRIBE.

5. Run DROP statements only when cleanup is required; they delete the objects.

Result: Hive commands were used to create a database and table, display them, and inspect the table structure.

Quick viva points

• CREATE DATABASE college; creates a database named college.

• USE college; makes college the current database.

- STRING is a Hive data type for text.
- DROP TABLE and DROP DATABASE delete objects; do not run them before checking the output.

Experiment 2: Vector Arithmetic in R

Aim: To perform addition, subtraction, multiplication, and division on two vectors.

Concept: In R, c() combines values into a vector. Arithmetic operations are applied element by element when vectors have matching lengths.

Program / Commands

```
a <- c(10, 20, 30)
b <- c(2, 4, 5)

print(a + b)
print(a - b)
print(a * b)
print(a / b)
```

Expected Output

```
[1] 12 24 35
[1]  8 16 25
[1]  20  80 150
[1] 5 5 6
```

Procedure

1. Open RStudio or the R console used in your lab.

2. Create a new R script, type the code, and run it.

3. Read the four output lines in order: addition, subtraction, multiplication, division.

Result: The four arithmetic operations were performed successfully on the vectors.

Quick viva points

• c() creates a vector.

• <- is the assignment operator.

• Vector arithmetic is element-wise.

• Division by zero needs special care in R.

Experiment 3: Sorting Vector Values

Aim: To sort the values 33, 55, 20, 34, 99, 23, 67, 82 in ascending and descending order.

Concept: sort(x) arranges values in increasing order. Setting decreasing = TRUE arranges them in decreasing order.

Program / Commands

```
x <- c(33, 55, 20, 34, 99, 23, 67, 82)

# Ascending order
print(sort(x))

# Descending order
print(sort(x, decreasing = TRUE))
```

Expected Output

```
[1] 20 23 33 34 55 67 82 99
[1] 99 82 67 55 34 33 23 20
```

Procedure

1. Open an R script or R console.

2. Create vector x using c().

3. Run sort(x) for ascending order.

4. Run sort(x, decreasing = TRUE) for descending order.

Result: The vector was sorted in ascending and descending order.

Quick viva points

• Ascending means smallest to largest.

• Descending means largest to smallest.

• The original vector x is not changed by sort() unless the sorted result is assigned back to x.

Experiment 4: Functions With and Without Arguments

Aim: To demonstrate a function without arguments and a function with arguments in R.

Concept: A function is defined using function(). Arguments pass input values into a function, and return() can send a result back to the caller.

Program / Commands

```
# Function without arguments
greet <- function() {
  print("Hello Students")
}
greet()

# Function with arguments
add <- function(a, b) {
  return(a + b)
}
result <- add(10, 20)
print(result)
```

Expected Output

```
[1] "Hello Students"
[1] 30
```

Procedure

1. Define greet with an empty argument list and call greet().

2. Define add with two arguments, a and b.

3. Call add(10, 20), store the result, and print it.

Result: Functions with and without arguments were defined and executed successfully.

Quick viva points

• function() defines a function.

• greet() calls the function without passing arguments.

• add(10, 20) passes 10 and 20 as arguments.

• return(a + b) returns the sum.

Experiment 5: Sequence, Mean, Maximum, and Sum

Aim: To create a sequence from 20 to 60 and calculate its mean, maximum, and sum.

Concept: seq(20, 60) generates integers from 20 through 60, inclusive. mean(), max(), and sum() calculate the average, largest value, and total.

Program / Commands

```
x <- seq(20, 60)

print(x)
print(mean(x))
print(max(x))
print(sum(x))
```

Expected Output

```
[1] 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40
[22] 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60
[1] 40
[1] 60
[1] 1640
```

Procedure

1. Open an R script or R console.

2. Create x using seq(20, 60).

3. Run print(x) to display the sequence.

4. Use mean(x), max(x), and sum(x) to calculate the required statistics.

Result: The sequence was created and its mean, maximum, and sum were calculated successfully.

Quick viva points

• The sequence includes both 20 and 60, so it contains 41 numbers.

• The mean is 40.

• The maximum is 60.

• The sum is 1640.

Quick Revision: Common Commands

R expression	Use
c(1, 2, 3)	Create a vector
sort(x)	Sort ascending
sort(x, decreasing = TRUE)	Sort descending
function(a, b) { ... }	Define a function with arguments
seq(20, 60)	Generate a sequence
mean(x) / max(x) / sum(x)	Calculate average / maximum / total
CREATE / SHOW / USE / DESCRIBE / DROP	Hive database and table commands

Note: Hive output and startup commands can differ by Hadoop/Hive version and lab configuration. Use your instructor's configured interface if it differs from the examples.